

MATH 20700 Honors Analysis I

Homework 3

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Problem 2

Proof. a) For uniform continuity, we want a single delta (dependent on epsilon) that works for all points $u, x \in (a, b)$.

We want $H|u - x|^\alpha < \epsilon$ where $|u - x| \leq \delta$, so it is clear that picking $\delta \leq \sqrt[\alpha]{\frac{\epsilon}{H}}$ proves uniform continuity.

Clearly, we can infer that the α -Hölder function $f : (a, b) \rightarrow \mathbb{R}$ extends uniquely to a continuous function $g : [a, b] \rightarrow \mathbb{R}$, namely such that

$$g(y) = \begin{cases} \lim_{x \rightarrow a} f(x) & \text{if } y = a \\ f(y) & \text{if } a < y < b \\ \lim_{x \rightarrow b} f(x) & \text{if } y = b. \end{cases}$$

We claim that the extended function is indeed α -Hölder. If $u, x \in (a, b)$ then we are done.

Case 1: $u \in \{a, b\}$ and $x \in (a, b)$, i.e., one of the points is an end point. Without loss of generality suppose $u = a$ and $x \in (a, b)$.

Consider a sequence of points (u_n) in the interval (a, b) such that $u_n \rightarrow a$, i.e., $\lim_{n \rightarrow \infty} u_n = a$. Then, $|f(a) - f(x)| = \lim_{n \rightarrow \infty} |f(u_n) - f(x)| \leq \lim_{n \rightarrow \infty} H|u_n - x|^\alpha = H|a - x|^\alpha$. The case $u = b$ is analogous.

Case 2: if $u, x \in \{a, b\}$, i.e., both of the points are end points. We do the same as above, except we now take two sequences $(p_n), (q_n)$ of points in (a, b) such that $p_n \rightarrow a$ and $q_n \rightarrow b$.

Then, $|f(a) - f(b)| = \lim_{n \rightarrow \infty} |f(p_n) - f(q_n)| \leq \lim_{n \rightarrow \infty} H|p_n - q_n|^\alpha = H|a - b|^\alpha$. Hence, the extended function is α -Hölder.

b) It means the function is Lipschitz continuous.

c) Since $|f(x) - f(u)| \leq H|x - u|^\alpha$, we see that $\frac{|f(x) - f(u)|}{|x - u|} \leq H|x - u|^{\alpha-1}$ and $\alpha - 1 > 0$. Taking the limit $u \rightarrow x$, we see that $\lim_{x \rightarrow u} \frac{|f(x) - f(u)|}{|x - u|} \leq \lim_{x \rightarrow u} H|x - u|^{\alpha-1} = 0$. Hence $f'(x) = 0$ for all x (since x was arbitrary).

QED

Problem 5

Proof. Yes, it follows that $f'(0)$ exists.

To prove this, we must show that the limit defining the derivative at 0 exists and equals L . By definition, we need to show

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = L.$$

Consider the interval $[0, h]$ (if $h > 0$) or $[h, 0]$ (if $h < 0$). By the Mean Value Theorem, there exists a number c_h strictly between 0 and h such that

$$f'(c_h) = \frac{f(h) - f(0)}{h - 0} = \frac{f(h) - f(0)}{h}.$$

Now, we need to find the limit of this expression as $h \rightarrow 0$. As $h \rightarrow 0$, the value c_h is “squeezed” between 0 and h . By the Squeeze Theorem, we must have $\lim_{h \rightarrow 0} c_h = 0$.

We can now take the limit of both sides of our equation from the MVT:

$$\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} f'(c_h)$$

Since we know $\lim_{h \rightarrow 0} c_h = 0$ and we are given that $\lim_{x \rightarrow 0} f'(x) = L$, it follows that

$$\lim_{h \rightarrow 0} f'(c_h) = \lim_{c_h \rightarrow 0} f'(c_h) = L$$

Therefore, we have shown

$$\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = L$$

This proves by definition that $f'(0)$ exists and $f'(0) = L$.

QED

Problem 9

Proof. a) We define $g(x) = f(x) - x$, then we want to show that $\exists! y \in \mathbb{R}$ such that $g(y) = 0$.

For existence, let $k = L - 1$, then we have $g'(x) < k < 0$.

By the Mean Value Theorem on the interval $[0, x]$ for any $x > 0$, there exists a $c \in (0, x)$ such that

$$\frac{g(x) - g(0)}{x - 0} = g'(c)$$

$$g(x) = g(0) + g'(c) \cdot x$$

Since $g'(c) < k$ and $x > 0$, we have $g'(c) \cdot x < kx$. Therefore:

$$g(x) < g(0) + kx$$

As $x \rightarrow \infty$, $g(0)$ is a constant and k is a negative constant, so

$$\lim_{x \rightarrow \infty} (g(0) + kx) = -\infty$$

Thus, $\lim_{x \rightarrow \infty} g(x) = -\infty$. Analogously for the other case of $[x, 0]$ for any $x < 0$, we see that $\lim_{x \rightarrow -\infty} g(x) = \infty$. Then, by Intermediate Value Theorem, $\exists y \in \mathbb{R}$ such that $g(y) = 0$.

For uniqueness, suppose for sake of contradiction that we have $z \neq y$ and $g(z) = g(y) = 0$. Without loss of generality suppose $y < z$. We note that $g'(x) = f'(x) - 1 < L - 1 < 0$ hence g is strictly decreasing, so $y < z$ implies $0 = g(y) > g(z) = 0$ which is a contradiction.

b) Consider $f(x) = x + e^{-x}$, then $f'(x) = 1 - e^{-x} \rightarrow_{x \rightarrow \infty} 1$.

Setting $f(x) = x$, we see that $x + e^{-x} = x$ hence $e^{-x} = 0$ which has no solutions, hence the function f has no fixed points.

QED

Problem 11

Proof. a) We know that $f''(x)$ exists, hence $f'(y)$ exists in an open interval around x . The limit is in the form $\frac{0}{0}$, so we can apply L'Hôpital's Rule by differentiating the numerator and denominator with respect to h .

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} &\stackrel{L'H}{=} \lim_{h \rightarrow 0} \frac{\frac{d}{dh}[f(x+h) - 2f(x) + f(x-h)]}{\frac{d}{dh}[h^2]} \\ &= \lim_{h \rightarrow 0} \frac{f'(x+h) \cdot 1 - 0 + f'(x-h) \cdot (-1)}{2h} \\ &= \lim_{h \rightarrow 0} \frac{f'(x+h) - f'(x-h)}{2h} \end{aligned}$$

Then, we note that

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f'(x+h) - f'(x-h)}{2h} &= \lim_{h \rightarrow 0} \frac{1}{2} \left[\frac{f'(x+h) - f'(x) + f'(x) - f'(x-h)}{h} \right] \\ &= \frac{1}{2} \left[\lim_{h \rightarrow 0} \frac{f'(x+h) - f'(x)}{h} + \lim_{h \rightarrow 0} \frac{f'(x) - f'(x-h)}{h} \right] \end{aligned}$$

We evaluate the two limits separately. For the first limit, $\lim_{h \rightarrow 0} \frac{f'(x+h) - f'(x)}{h} = f''(x)$ (by definition of the derivative).

For the second limit, let $k = -h$. As $h \rightarrow 0$, $k \rightarrow 0$.

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f'(x) - f'(x-h)}{h} &= \lim_{k \rightarrow 0} \frac{f'(x) - f'(x+k)}{-k} \\ &= \lim_{k \rightarrow 0} \frac{-(f'(x+k) - f'(x))}{-k} \\ &= \lim_{k \rightarrow 0} \frac{f'(x+k) - f'(x)}{k} \\ &= f''(x). \end{aligned}$$

Substituting these results back into the equation:

$$\frac{1}{2}[f''(x) + f''(x)] = \frac{1}{2}[2f''(x)] = f''(x)$$

This completes the proof.

b) Consider the function $f(x) = x|x|$ at the point $x = 0$.

$f''(0)$ fails to exist. The function can be written as $f(x) = \begin{cases} x^2 & \text{if } x \geq 0 \\ -x^2 & \text{if } x < 0 \end{cases}$. The first derivative is $f'(x) = 2|x|$. We check the left-hand and right-hand limits for the derivative of $f'(x)$ at $x = 0$:

- Right-hand limit: $\lim_{h \rightarrow 0^+} \frac{f'(0+h) - f'(0)}{h} = \lim_{h \rightarrow 0^+} \frac{2h - 0}{h} = 2$
- Left-hand limit: $\lim_{h \rightarrow 0^-} \frac{f'(0+h) - f'(0)}{h} = \lim_{h \rightarrow 0^-} \frac{-2h - 0}{h} = -2$

Since the limits 2 and -2 are not equal, $f''(0)$ does not exist.

The symmetric limit exists. We compute the limit from part (a) at $x = 0$:

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(0+h) - 2f(0) + f(0-h)}{h^2} &= \lim_{h \rightarrow 0} \frac{f(h) - 0 + f(-h)}{h^2} \\ &= \lim_{h \rightarrow 0} \frac{h|h| + (-h)|-h|}{h^2} \\ &= \lim_{h \rightarrow 0} \frac{h|h| - h|h|}{h^2} \quad (\text{since } |-h| = |h|) \\ &= \lim_{h \rightarrow 0} \frac{0}{h^2} = \lim_{h \rightarrow 0} 0 = 0. \end{aligned}$$

Thus, the limit exists and is 0, even though $f''(0)$ does not exist.

QED